

STRATA

AI-Native Metals & Mining Intelligence Platform

Full Product Blueprint — Architecture · Personas · Use Cases · Business Model

CONFIDENTIAL | 2025

Competitive intelligence platform — next-generation Wood Mackenzie alternative

Battery Metals

AI Agents

Real-Time Data

SaaS

Contents

01 Executive Summary & Vision

02 Market Opportunity

03 User Personas

04 Core Use Cases

05 Full Application Architecture

06 Data Layer & Sources

07 AI Agent Architecture

08 Business Model & Pricing

09 Go-to-Market Strategy

10 Financial Projections

11 Build Roadmap

12 Team & Hiring Plan

13 Risk Factors

01 EXECUTIVE SUMMARY & VISION

What is Strata?

Strata is an AI-native metals and mining intelligence platform that replaces the dominant but outdated model of expensive analyst-authored reports with continuous agentic data collection, real-time AI synthesis, and a conversational interface. It targets the same buyer that today pays Wood Mackenzie \$80,000–\$300,000 per year for slow, static research — and delivers a superior product at a fraction of the cost.

The platform aggregates data from hundreds of public sources — stock exchange filings, government mining databases, exchange price feeds, satellite imagery, and shipping data — normalises and enriches it using large language model extraction pipelines, and makes it queryable through both a structured dashboard and a natural-language conversational interface. Users can upload their own private mine models and financial data to blend proprietary information with the public dataset.

\$80–300k	\$12–60k	Real-time	10×
WoodMac annual cost	Strata target price	Data freshness	Cost advantage
enterprise subscription	full access, all tiers	vs quarterly analyst cycle	at launch positioning

The Core Problem

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Data latency	Quarterly analyst publication cycle	Continuous agentic ingestion, updated daily
No interactivity	Read static PDFs and dashboards	Conversational AI interface over live data
High cost	\$80k–\$300k/yr enterprise licences	\$299–\$5k+/mo, usage-based tiers
No private data	Zero integration with internal models	Private data upload + blended queries
No alerting	Manual monitoring required	Agentic alert engine with AI explanations
Analyst bottleneck	Report published on analyst schedule	On-demand AI report generation in minutes

02 MARKET OPPORTUNITY

Wood Mackenzie was acquired by Verisk Analytics for \$2.8 billion in 2015 and generates estimated revenues of \$500M+ annually. S&P; Global Commodity Insights (formerly Platts) is a \$3B+ revenue business. The combined addressable market for commodity intelligence products is conservatively \$8–12B globally, with mining and metals representing approximately 20–25% of that base.

The battery metals segment — lithium, cobalt, nickel, manganese — represents the fastest-growing subsector by data demand, driven by the EV transition, government critical mineral strategies, and the rapid buildout of gigafactory supply chains. This segment is structurally underserved: Wood Mackenzie's battery metals coverage is relatively thin compared to its traditional base metals franchise, while demand from EV manufacturers, battery cell makers, and government agencies is exploding.

\$8–12B	\$1.8B	\$400M	22% CAGR
Total addressable market	Mining & metals segment	Battery metals segment	Battery data demand
commodity intelligence globally	estimated TAM	fastest-growing subsector	2023–2028 estimate

Strategic Wedge — Why Battery Metals First

A focused vertical entry is essential. Trying to replicate Wood Mackenzie's full commodity coverage from day one would require hundreds of millions in capital and years of data accumulation. Instead, Strata enters through battery metals for four reasons:

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Undercovered	WoodMac's battery metals franchise is relatively thin; specialist competitors (Benchmark Mineral Intelligence, Fastmarkets) are expensive and non-interactive.
Urgent demand	EV manufacturers and battery cell makers need real-time procurement intelligence. Their supply chain teams have large budgets and acute pain.
Public data richness	Battery metal miners (Albemarle, Livent, SQM, Ganfeng, Vale) are prolific public company disclosers. AISC data extraction is highly tractable.
Government buyers	Critical mineral strategies in the US, EU, Japan, and Australia have created well-funded government agency buyers who are priced out of WoodMac.

03 USER PERSONAS

Six primary personas with distinct workflows, data needs, willingness-to-pay, and product surface requirements. The portfolio manager and battery supply chain lead are the highest-value early adopters; the academic/researcher persona anchors the low-cost entry tier and drives top-of-funnel awareness.

Resource Analyst

Role	Mining company — corporate development or strategy
Pain point	Benchmarks asset costs vs peers, screens M&A; targets, builds supply models. Spends hours manually extracting AISC data from quarterly filings across 20+ companies.
Value delivered	Instant AI-powered peer benchmarking. Cost curve always current. Private mine model upload for internal vs external comparison.
Willingness to pay	\$500–2k/mo · Best fit: Professional or Enterprise

Portfolio Manager

Role	Commodity hedge fund or resource-focused long/short equity
Pain point	Tracks price catalysts, mine disruptions, geopolitical risk across a book of mining equities and commodity derivatives. Needs fast, sourced intelligence.
Value delivered	Real-time supply disruption feed. Agentic alerts on production guidance cuts. AI-synthesised market commentary on demand.
Willingness to pay	\$1–5k/mo · Best fit: Professional or Enterprise

Battery Supply Chain Lead

Role	EV manufacturer or battery cell maker (OEM/Tier 1)
Pain point	Forecasts raw material cost and availability 3–5 years out across Li, Ni, Co, Mn. Models procurement scenarios and monitors supplier concentration risk.
Value delivered	Demand scenario modelling. Supplier risk profiles. Price forecast engine with EV penetration scenario integration. Private procurement data upload.
Willingness to pay	\$2–10k/mo · Best fit: Enterprise

PE / Infrastructure Investor

Role	Private equity, infrastructure fund, or project finance bank
Pain point	Evaluates greenfield and brownfield mine investments. Needs commodity price scenarios, capex benchmarking, and regional risk profiles to anchor financial models.
Value delivered	AI-generated investment memos. Automated DD packs from NI 43-101 uploads. NPV scenario generator. Comparable transaction data.
Willingness to pay	\$2–5k/mo · Best fit: Professional or Enterprise

Government Policymaker

Role	Energy ministry, central bank, strategic reserve agency
Pain point	Tracks critical mineral supply security, import dependency ratios, and price volatility for strategic planning, tariff modelling, and stockpiling decisions.
Value delivered	Supply security dashboards. Import dependency metrics. Policy impact modelling. Affordable academic/gov tier pricing.
Willingness to pay	\$200–1k/mo · Best fit: Academic/Gov or Professional

Mining Researcher / Journalist

Role	Academic institution, think tank, or specialist press
Pain point	Needs sourced, citable production data, cost curves, and market structure analysis. Currently stitches USGS + company reports manually. Can't afford WoodMac.
Value delivered	Low-cost access tier with full data. Citation export mode. Research query interface. Historical data depth.
Willingness to pay	\$200/mo · Best fit: Explorer or Academic/Gov

04 CORE USE CASES

Twelve primary use cases organised across four workflow categories. Each maps to a specific product capability, one or more user personas, and a revenue tier.

Research & Intelligence

Use case	On-demand commodity reports	Personas	Resource Analyst, Portfolio Mgr
Description	User selects commodity, region, and timeframe. An agentic workflow fetches the latest supply/demand data, runs through the price model, and generates a structured research report with full citations. Replaces Wood Mackenzie's quarterly analyst PDF cycle with on-demand generation in under 60 seconds.		
Use case	Conversational data queries	Personas	All personas
Description	Natural-language interface over the full data corpus and structured database. Example: 'What is the marginal cost of Chilean lithium production given current brine grade decline rates and new water allocation restrictions?' The system retrieves relevant data, reasons over it, and returns a cited, structured answer.		
Use case	Regulatory & news monitoring	Personas	Supply Chain Lead, Portfolio Mgr
Description	Continuous agent monitors regulatory filings, mining ministry announcements, exchange position reports, and specialist financial news. Surfaces events with an AI-generated impact assessment: commodity affected, estimated volume displaced, and price direction.		
Use case	Geopolitical risk scoring	Personas	PE Investor, Government
Description	Country and asset-level risk scores updated from political event data, ESG incident feeds, and labour dispute monitoring. Produces a risk-adjusted supply outlook — showing what percentage of global production sits in high-risk jurisdictions per commodity.		

Cost & Asset Analytics

Use case	Interactive cost curve builder	Personas	Analyst, PE Investor
Description	Dynamic cost curves for any commodity, rendered from AI-extracted AISC/C1 data across all publicly reporting producers. Users stress-test input assumptions — energy price, labour cost, FX, strip ratio — and see the curve re-render in real time. Identifies high-cost producers at risk of curtailment at various price levels.		
Use case	Asset-level benchmarking	Personas	Mining company
Description	Users upload their mine's financial model or cost data (private, fully isolated). The platform benchmarks AISC, sustaining capex, and productivity metrics against the global peer set, showing percentile position on the industry cost curve with drill-down to individual cost components.		
Use case	Project NPV scenario generator	Personas	PE Investor, Analyst
Description	User inputs project parameters: reserves, grade, capex, opex, ramp schedule. The AI generates base/bull/bear NPV scenarios with commodity price assumptions sourced from Strata's own forecast model. Output is exportable to Excel with full assumption tables.		
Use case	Supply disruption tracker	Personas	Supply Chain Lead, Portfolio Mgr
Description	Live monitoring of mine curtailments, force majeure, and production guidance downgrades. Calculates displaced production as a percentage of global supply and models price impact across bear/base/bull demand scenarios in real time.		

Forecasting & Modelling

Use case	Price forecast engine	Personas	All personas
Description	Quantitative supply/demand model continuously updated from ingested data. Shows 1/3/5-year price forecasts with confidence intervals per commodity. Users can override any assumption and see the model re-calculate live. Tracks forecast vs realised performance over time.		
Use case	Demand scenario modelling	Personas	Supply Chain Lead, Government
Description	Models EV penetration curves, grid storage deployment, and green hydrogen demand scenarios and their impact on battery metal demand. Integrates IEA, BloombergNEF, and proprietary demand projections as selectable scenario sets.		

Agentic Alerts & Workflows

Use case	Personalised alert engine	Personas	Portfolio Mgr, Supply Chain Lead
Description	Users configure threshold and event-based alerts in natural language: 'Alert me when lithium carbonate spot price diverges more than 15% from our forecast', or 'Notify me when any DRC cobalt mine reports a force majeure'. The agent monitors continuously and delivers notifications via email, Slack, or webhook with plain-language context.		
Use case	Automated due diligence packs	Personas	PE Investor, Analyst
Description	User uploads target asset documents: NI 43-101 technical report, feasibility study, historical financials, management presentations. The AI cross-references against Strata's asset database and produces a structured DD summary covering resource quality, cost position, jurisdiction risk, and comparable transaction benchmarks.		

05 FULL APPLICATION ARCHITECTURE

A seven-layer stack from raw data ingestion through to user-facing interfaces. Each layer is independently scalable and deployable. The stack is designed to be cloud-native from day one, with horizontal scaling on the ingestion and AI layers as data volume grows.

Layer 1	Data Ingestion Layer — Agentic Crawlers
Description	Scheduled and event-triggered Python agents collect from public filings (EDGAR, SEDAR, ASX, JSE, TSX), government databases (USGS, World Bank, national mining ministries), exchange price APIs (LME, CME, SHFE), AIS vessel tracking, satellite change-detection feeds, and financial news APIs. Each agent is specialised per source type with its own error handling, retry logic, and output schema. A Kafka event bus decouples ingestion from downstream processing.
Tech stack	Python · Playwright · BeautifulSoup · APScheduler · Kafka · AIS API · Sentinel-2 / Planet Labs
Layer 2	Data Normalisation & Enrichment
Description	Raw data is cleaned, normalised, and semantically enriched. An LLM-powered extraction pipeline pulls structured cost components from unstructured annual reports and technical documents. A knowledge graph maps relationships between companies, assets (mines, processing plants, refineries), regions, and commodities. Entity resolution deduplicates companies and assets across sources using fuzzy matching and LLM disambiguation.
Tech stack	Claude API (extraction) · dbt · Apache Spark · Neo4j · OpenStreetMap · Dedupe.io
Layer 3	Storage Layer — Multi-Model
Description	Time-series commodity price data in TimescaleDB (a Postgres extension). Structured asset and company master data in PostgreSQL with row-level security for multi-tenancy. Document corpus — filings, reports, news articles — in a hybrid vector database (Pinecone) for semantic retrieval with BM25 keyword fallback. Knowledge graph in Neo4j. Geospatial mine and deposit data in PostGIS. All raw source files archived in S3.
Tech stack	PostgreSQL · TimescaleDB · Pinecone (vectors) · Neo4j · PostGIS · Amazon S3
Layer 4	AI Orchestration — Agents + RAG
Description	The core intelligence layer. A multi-agent system built on LangGraph with specialised sub-agents for each task type: price forecasting, filing extraction, news synthesis, alert evaluation, and report generation. A supervisor agent routes incoming queries, decomposes them into sub-tasks, fans out to specialist agents, and assembles cited responses. Retrieval is hybrid: semantic (vector similarity), keyword (BM25), and structured (SQL) — the retrieval agent selects the appropriate strategy per query type and reranks results before synthesis.
Tech stack	Claude API (claude-sonnet-4) · LangGraph · pgvector + Pinecone · MCP tool servers · LangSmith
Layer 5	Quantitative Models Layer
Description	Parameterised supply/demand models per commodity, maintained as versioned Python modules. The cost curve engine re-renders dynamically as input data updates from the ingestion layer. Monte Carlo simulation for price scenarios with configurable input distributions. All model assumptions are transparent and user-overridable via the frontend. Model outputs are cached in TimescaleDB and invalidated when underlying data changes.
Tech stack	Python · NumPy · SciPy · Pandas · Statsmodels · custom S/D model modules · FastAPI model server
Layer 6	API & Backend Services

Description	A FastAPI application server exposes both REST and GraphQL endpoints for all platform data. A Celery worker pool handles background jobs: agentic alert evaluation (every 15 minutes), scheduled report generation, data refresh orchestration, and file processing for private data uploads. Authentication and authorisation via Auth0 with SAML/SSO support for enterprise. Multi-tenancy enforced at the database layer using Postgres row-level security; private tenant data is never accessible across tenant boundaries.
Tech stack	FastAPI · GraphQL (Strawberry) · Celery · Redis (queue + cache) · Auth0 · Postgres RLS
Layer 7	Frontend — Dashboard + Conversational UI
Description	A React single-page application with three primary surfaces: (1) a data dashboard with interactive charts, cost curves, and real-time price feeds; (2) a conversational research interface — a chat-over-data UI with inline citations linking to source data; and (3) an alerts and workflow management console for configuring and reviewing agentic alert conditions. WebSockets deliver live price feed updates. A white-label configuration option is available for enterprise customers. Mobile-responsive.
Tech stack	React · TypeScript · Recharts / D3 · Tailwind CSS · Vercel · WebSockets

Infrastructure & DevOps

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Cloud provider	AWS (primary)	ECS (containerised agents), RDS (Postgres), ElastiCache (Redis), S3. Multi-region from Year 2.
Containerisation	Docker + ECS Fargate	Serverless containers for ingestion agents — scales to zero, auto-scales on queue depth.
CI/CD	GitHub Actions	Automated testing, staging deployment, and production release pipeline.
Monitoring	Datadog + LangSmith	Datadog for infra/APM; LangSmith for LLM trace logging, prompt performance, and agent debugging.
Error tracking	Sentry	Frontend and backend error capture with release tracking.
Security	SOC 2 Type II	Achieved in Year 1. Critical for enterprise sales. Pen testing quarterly.
Secrets management	AWS Secrets Manager	All API keys, DB credentials, and LLM API tokens rotated automatically.

06 DATA LAYER & SOURCES

Every data source used at launch is either free, low-cost, or accessible via a sub-\$5,000/yr API subscription. The competitive moat is not the raw source — it is the collection infrastructure, normalisation quality, AI extraction accuracy, and historical depth accumulated over time.

Price & Exchange Data

Source	Description & Access
LME (London Metal Exchange)	Free delayed API; real-time via Quandl or direct LME subscription (~\$2k/yr). Base metals spot and futures: copper, nickel, zinc, aluminium, lead, tin.
CME Group	CME DataMine API. Copper, gold, silver futures. Critical for US market pricing and options data.
SHFE (Shanghai Futures Exchange)	Chinese domestic metal prices. Essential for understanding the China premium/discount and arbitrage signals.
Fastmarkets / Benchmark Mineral Intelligence	Battery-grade lithium carbonate, cobalt sulphate, nickel sulphate spot assessments. ~\$5–15k/yr subscription. Most critical paid data source for Phase 1.
World Bank Commodity Price Data	Free long-run price series (30+ years) for all major commodities. Essential for historical model calibration and backtesting.
Quandl / Nasdaq Data Link	Aggregated commodity price APIs with clean, standardised formats. ~\$500/mo for relevant commodity data packages.

Production & Company Filings

Source	Description & Access
EDGAR (SEC)	Free API. US-listed mining company 10-K, 10-Q, 8-K filings. Primary source for AISC disclosure from Freeport, Albemarle, Livent, MP Materials, and others.
SEDAR+ (Canada)	Canadian securities filings. Covers TSX-listed miners: Barrick, Teck, First Quantum, Lithium Americas, and hundreds of junior explorers.
ASX Announcements	Australian Securities Exchange continuous disclosure. Covers Pilbara Minerals, Allkem, IGO, BHP, Rio Tinto lithium/nickel assets.
NI 43-101 / JORC Technical Reports	Resource/reserve technical documents. AI extracts ore grade, strip ratio, recovery rates, capital cost estimates, and mine life from PDF. Filed on SEDAR and ASX.
National Mining Ministries	Chile (SERNAGEOMIN), DRC (CAMI), Australia (Geoscience Australia), Indonesia (ESDM). Country-level production permit databases and official production statistics.
USGS Mineral Resources Program	Free. Annual Mineral Commodity Summaries — the authoritative source for global production volumes by country and commodity.

Alternative & Signal Data

Source	Description & Access
Sentinel-2 Satellite Imagery (ESA)	Free, 10m resolution, 5-day revisit. Used for mine activity monitoring: tailings dam growth, stockpile change detection, construction progress at greenfield sites.
Planet Labs	Commercial upgrade for higher-resolution, daily imagery at selected high-priority assets. ~\$30k/yr for a targeted asset list. Layer 2 investment.

AIS Vessel Tracking	MarineTraffic API (~\$2k/mo). Shipping route intelligence for ore and refined metal trade flows. Port-level loading/unloading signals for export volumes.
Factiva / Dow Jones News API	Mining-specific news filtering. Labour disputes, permit approvals, force majeure, ESG incidents. ~\$5k/yr with usage-based components.
S&P; Global Commodity Insights RSS	Free RSS feeds for spot market news and production alerts. Supplemented by Mining.com and Kitco for base and precious metals.
OpenStreetMap + Overpass API	Free geospatial infrastructure: road access, port proximity, power grid. Used for logistics cost modelling in the cost curve engine.

07 AI AGENT ARCHITECTURE

A supervisor-worker multi-agent system built on LangGraph. Each specialist agent has a defined scope, tool access, prompt template, and output schema. The supervisor agent handles query routing, task decomposition, context assembly, and final response synthesis. All agent traces are logged to LangSmith for debugging and prompt performance monitoring.

Agent	Supervisor Agent
Function	Entry point for all user queries. Classifies intent (research query / data lookup / alert configuration / report request / private data question). Decomposes complex queries into parallel sub-tasks. Routes to specialist agents. Assembles multi-source responses with inline citations. Manages clarification turns when query is ambiguous.
Implementation	claude-sonnet-4 · LangGraph StateGraph · intent classification · task decomposition
Agent	Retrieval Agent (RAG)
Function	Hybrid retrieval across three stores: (1) vector store — semantic search over filings and reports using Pinecone; (2) structured store — SQL queries over the PostgreSQL asset and price database; (3) knowledge graph — Cypher queries over Neo4j for relationship traversal (e.g. 'all assets owned by companies in the DRC'). Applies a learned reranker to merge results before returning context to the supervisor.
Implementation	Pinecone + pgvector · PostgreSQL · Neo4j (Cypher) · Cohere reranker
Agent	Filing Extraction Agent
Function	Specialised in parsing mining company filings using a chain-of-thought extraction protocol. Handles inconsistent cost reporting formats across jurisdictions (AISC vs C1 vs AISC+sustaining). Extracts: production volumes, unit cost components (energy, labour, reagents, royalties), capex guidance, and forward guidance language. Runs continuously as a background job against the filing ingestion queue.
Implementation	claude-sonnet-4 · PDF text extraction · structured output schema · Celery worker
Agent	Price Forecast Agent
Function	Interfaces with the quantitative supply/demand model layer via a Python REPL tool. Accepts user-defined scenario overrides (EV penetration rate, Chinese smelter capacity, energy costs). Re-runs the model with new parameters and explains the delta vs base case in plain language. Flags when spot price is trading significantly above or below model fair value with a cited rationale.
Implementation	claude-sonnet-4 · Python REPL tool · S/D model API · TimescaleDB
Agent	Alert Evaluation Agent
Function	Background agent that runs every 15 minutes against all subscribed user alert conditions. For threshold alerts: evaluates current data vs configured thresholds. For event alerts: scans recent news and filing events for pattern matches. When triggered, generates a structured notification: what happened, supporting data evidence, magnitude/severity, and suggested next steps for the user.
Implementation	Celery scheduled task · event stream consumer · claude-sonnet-4 · email/Slack/webhook
Agent	Report Generation Agent
Function	On-demand or scheduled structured report creation following a configurable template: executive summary, supply analysis, demand outlook, price forecast, risk factors, and appendices with data tables. Every factual claim is grounded with a citation to a specific source document or data point in the system. Output is rendered as a styled PDF or DOCX. Reports are versioned and stored per user.

Private Data Integration

Private data integration is the most defensible enterprise feature. Users upload their own mine financial models, portfolio spreadsheets, internal cost assumptions, and board pack PDFs. Data is stored in a fully isolated tenant partition, never used to train or fine-tune models, never accessible across tenant boundaries, and blended into AI responses only when the query context makes it relevant. The blended query capability — 'How does our Atacama asset's AISC compare to the industry cost curve?' — is not available in any competing product.

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Supported formats	Excel (.xlsx), CSV, PDF (technical reports, NI 43-101, board packs), structured JSON via REST API.
Isolation model	Row-level security in Postgres. Separate vector namespaces per tenant in Pinecone. Encryption at rest with per-tenant KMS keys.
Data retention policy	Configurable. Default 90-day retention with explicit extension. Data can be deleted on demand via API or UI.
Blended queries	Private data is injected into retrieval context only when the supervisor agent determines it is relevant to the query. Never surfaced to other tenants.
Audit logging	All private data accesses are logged with user ID, timestamp, and query context. Available to enterprise admins via the management console.

08 BUSINESS MODEL & PRICING

SaaS subscription is the primary revenue engine, complemented by three secondary streams that emerge naturally from the platform's capabilities: a metered data API, custom research projects, and bulk data licensing to financial data vendors.

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Price	\$299/mo	\$999/mo	\$5,000+/mo	\$199/mo
Target user	Individual analysts, researchers	Professional teams (2–10 seats)	Mining firms, funds, OEMs	Universities, ministries
Commodity coverage	1 vertical	All battery metals	All commodities	All battery metals
AI queries	50/month	Unlimited	Unlimited	200/month
Cost curve access	Read-only	Full + stress-test	Full + custom inputs	Read-only
Alert engine	Community alerts only	Custom threshold + event alerts	Custom + team alerts + webhooks	Community alerts
Private data upload	No	5 GB	Unlimited	No
Report generation	No	20 reports/month	Unlimited + custom templates	10 reports/month
API access	No	Limited (1k calls/mo)	Full API + bulk export	Limited (500 calls/mo)
SSO / User management	No	No	Yes (SAML/OIDC)	No
SLA / Support	Email support	Priority email	Dedicated CSM + 99.9% SLA	Email support

Secondary Revenue Streams

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Data API (metered)	Metered REST API for price feeds, cost curve data, and production data. Targets quant funds, Bloomberg terminal developers, and software vendors.	\$0.01–\$0.05/call or \$500–5k/mo bulk packages	Year 2
Custom research	AI-assisted bespoke research projects: M&A; due diligence, greenfield assessments, regulatory impact analysis. Delivered as consulting-style engagements.	\$10,000–\$50,000/project	Year 1
Data licensing	Sell normalised, AI-enriched mining asset data as bulk datasets to Bloomberg, Refinitiv, FactSet — who today licence from WoodMac at high margins.	\$50,000–\$500,000/yr enterprise deals	Year 2–3

09 GO-TO-MARKET STRATEGY

A content-led, product-led growth motion. The platform's primary GTM asset is a free weekly battery metals intelligence briefing powered by the Strata platform itself — demonstrating the product's capability to the exact audience that would buy it. Paid acquisition is minimal until product-market fit is confirmed at ~50 paying customers.

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Content-led inbound	Free weekly battery metals briefing (email + LinkedIn). Published using Strata's own agents to demonstrate the product.	Resource analysts, portfolio managers, supply chain leads following commodity markets	Email subscribers → Explorer trial conversion
Product-led growth	Explorer tier at \$299/mo removes friction. Converts to Professional when query limits hit or private data needed.	Individual analysts at target firms	Explorer → Pro upgrade rate (target: 25%)
Top-down enterprise	Outbound to VP Supply Chain / VP Strategy at EV OEMs and battery cell makers. Start with a custom research project as a paid pilot.	EV manufacturers, battery cell makers, mining strategy teams	Enterprise ACV > \$60k/yr
Community & events	Sponsor / present at Mining Indaba, PDAC, and battery materials conferences. Host a free annual battery metals outlook webinar.	Senior industry decision makers	Pipeline generated per event
Partner channel	Integrate with financial data platforms (Bloomberg API, Refinitiv Workspace) to distribute data. White-label for industry consultancies.	Bloomberg / Refinitiv enterprise users	Data licensing revenue

10 FINANCIAL PROJECTIONS

Conservative projections based on product-led growth motion with content-led top-of-funnel. Year 1 is pre-revenue or early revenue; Series A is targeted for end of Year 2 / early Year 3 based on \$3–5M ARR milestone.

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Explorer subs (\$299/mo)	20	80	200	400
Professional subs (\$999/mo)	15	120	400	800
Enterprise contracts (\$5k+/mo avg \$8k)	1	8	25	60
API / data licensing revenue	\$0	\$120k	\$600k	\$1.8M
Custom research revenue	\$50k	\$200k	\$500k	\$800k
Total ARR	~\$250k	~\$1.6M	~\$6.5M	~\$16M
Gross margin	60%	68%	74%	78%
Team size	4	10	22	38
Monthly burn (pre-Series A)	\$80k	\$150k	—	—

Unit Economics Targets (Year 3 Steady State)

\$16k	<\$8k	2.0×	120%
Average contract value	Target CAC	LTV/CAC ratio Year 1	Target NRR
blended across all tiers	content-led, PLG motion	improving to 5× by Year 3	expansion via commodity add-ons

11 BUILD ROADMAP — 24 MONTHS

Phased build designed to generate first revenue at month 5–6, enterprise contracts by month 9, and Series A fundraise conditions by month 14. Each phase delivers a shippable, revenue-generating product — not just infrastructure.

Months 1–2	Foundation — Data Infrastructure + MVP AI
Deliverables	• LME/CME price ingestion pipeline operational • EDGAR + SEDAR filing crawlers with structured output • USGS + World Bank data feeds integrated • PostgreSQL + TimescaleDB schema designed and deployed • Pinecone vector store set up with basic RAG over filing corpus • Basic React dashboard: price charts + filing browser • MVP conversational query (single-turn, battery metals only)
Months 3–4	Alpha — Battery Metals Vertical + Cost Curves
Deliverables	• Full Li/Co/Ni/Mn coverage across all major public producers • Filing extraction agent v1: AISC and C1 cost extraction • Cost curve MVP: interactive, 20 major producers • Alert engine v1: threshold-based price alerts • Multi-turn conversational query with citations • 10 paying alpha customers (target: analysts at mining firms or funds)
Months 5–6	Beta Launch — Private Data + Report Generation
Deliverables	• Private data upload: Excel, CSV, PDF (fully tenant-isolated) • Blended private/public query capability live • Report generation agent: on-demand PDF report with citations • Stripe billing integrated, all subscription tiers live • Public beta announcement, targeting 50+ paying customers • Content flywheel: weekly battery metals briefing launched
Months 7–9	Enterprise Track + Copper & Zinc Expansion
Deliverables	• Copper and zinc coverage added (Phase 2 start) • SSO (SAML) and team/user management for enterprise • Price forecast model v1: 1-year forward curve with assumptions UI • Satellite imagery signals (Sentinel-2) integrated for top 20 assets • AIS vessel tracking for battery metal trade flows • First 3 enterprise contracts signed (target ACV \$60k+)
Months 10–14	Scale — API Product + Knowledge Graph
Deliverables	• Public data API launched with metered billing • Knowledge graph v1: asset-company-region relationship traversal • Due diligence pack feature: NI 43-101 / JORC AI analysis • NPV scenario generator with Excel export • Geopolitical risk scoring dashboard • Series A fundraise (\$5–10M target at ~\$3M ARR run-rate)
Months 15–24	Full Coverage — WoodMac Parity + Data Licensing

Deliverables	• Gold, silver, iron ore, coking coal commodity coverage • Potash, phosphate, rare earths added • Full supply chain graph (mine → refinery → battery → OEM) • White-label enterprise offering launched • Bloomberg / Refinitiv data licensing deal(s) signed • Full multi-region deployment (EU data residency for European enterprise)
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12 TEAM & HIRING PLAN

The founding team composition is the single most important determinant of early success. The domain co-founder — someone with existing relationships in battery supply chain or mining finance — is the critical hire. They close the first 10 alpha customers before the product is fully built and validate that the product solves a real, budgeted problem.

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Technical co-founder	Backend + AI engineering: Python, LLM APIs, data pipelines, cloud infrastructure. Has built data-intensive SaaS before.	Day 0	Owns the entire technical stack. Without this hire there is no product.

Domain co-founder	5–10 yrs in commodity research, mining finance, or battery supply chain. Has a Rolodex of the exact buyer persona. Can write the first weekly briefing.	Day 0	Closes first 10 alpha customers. Validates product direction. Is the credibility anchor for enterprise sales.
Frontend engineer	React + TypeScript + data visualisation (Recharts, D3). Has built dashboards before. Ideally some fintech/data product experience.	Month 2	Unblocks the user-facing product. Technical co-founder should not be building React.
Data engineer	Python, Airflow/Celery, SQL, cloud data infra. Experience building reliable web scraping and ETL pipelines.	Month 4	Data ingestion reliability is existential. Manual fixes to crawlers at scale don't work.
Commodity analyst	Junior analyst from a mining bank, WoodMac itself, or a specialist consultancy. Validates AI extraction quality and writes content.	Month 5	QA layer on AI-extracted cost data. Also writes the weekly briefing content that drives top-of-funnel.
Enterprise sales lead	Has sold B2B SaaS or data products to financial services or commodity firms. Comfortable with \$50k+ ACV sales cycles.	Month 8	The founding team can close first 5–10 customers. Cannot close 50+ without dedicated sales.
Customer success	Technical enough to onboard enterprise users, explain the AI, and expand usage. Mining or finance background a plus.	Month 10	NRR is the growth engine. Customer success owns the expansion from 100% to 120% NRR.
ML engineer	LLM fine-tuning, retrieval optimisation, agent evaluation. Has worked on RAG systems in production.	Month 12	Needed to systematically improve extraction accuracy and retrieval quality at scale.

13 RISK FACTORS & MITIGATIONS

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AI hallucination in cost data	High	Every factual claim in AI outputs is linked to a source document or database record. Red-teaming during alpha. Commodity analyst QA layer on extraction pipeline output.
Data source changes / scraping breaks	Medium	Redundant data sources per commodity. Automated monitoring on ingestion pipeline health. Fallback to cached data with staleness indicators.
Wood Mackenzie launches a chat interface	Medium	Their moat is analyst depth, not tech. A chat layer on static data doesn't give real-time or private data integration. Our differentiation is architectural, not UX.
LLM API cost at scale	Medium	Cost-per-query optimisation: cache common queries, use smaller models for classification tasks, batch extraction jobs. Target <\$0.10/query average at scale.
Data licensing / IP disputes	Low-Medium	All sources are public filings and free government data. Legal review of data collection practices in Year 1. Avoid scraping behind paywalls.
Enterprise sales cycle length	Medium	Mitigate with bottom-up PLG: individual users adopt Explorer/Pro, pull enterprise procurement. Reduces cold outbound dependency.
Key person dependency (domain co-founder)	High early	Hire commodity analyst by Month 5 to distribute domain knowledge. Document all source validation logic explicitly.
Commodity price data accuracy	High	Multiple source cross-validation for all price series. Automated anomaly detection on time series. Manual review alert when deviation > 2 standard deviations.